

Quantization for Closed-Loop Success on Edge Embedded Hardware: A Cross-Embodiment Study of Driving and Manipulation Vision–Language–Action Policies on the Jetson Orin

Justin Williams¹, Kishor Datta Gupta², Roy George³, and Mrinmoy Sarkar⁴

Department of Cyber-Physical Systems, Clark Atlanta University, Atlanta, GA, USA

¹justin.williams1@students.cau.edu, ²kgupta@cau.edu, ³rgeorge@cau.edu

Siemens Corporation, Princeton, NJ, USA

⁴mrinmoy.sarkar@siemens.com

Abstract—Vision–language–action (VLA) policies are moving from the lab to embedded robot and vehicle compute, where their large language backbones must be quantized to meet latency, power, and shared-memory budgets. Yet quantization is almost always validated on in-distribution, open-loop accuracy—which we show is a poor predictor of what the policy actually does when it is closing the loop on real hardware. We present a cross-embodiment study of post-training quantization for closed-loop success on the edge, spanning two very different VLA policies deployed on the same NVIDIA Jetson AGX Orin: a driving policy (OpenDriveVLA, an autoregressive-text trajectory head on nuScenes/NeuroNCAP) and a manipulation policy (OpenVLA-OFT, an L1-regression action head on LIBERO). The comparison exposes a unified picture. There are two distinct quantization failure modes, distinguished by when they trigger, not by the robot: (i) weight-scale granularity, which fails only under deployment-domain shift, and (ii) activation outliers, which fail universally and in-distribution. Each has a near-free fix—group-wise weight scales and an orthogonal activation rotation, respectively—and we show the fixes transfer across embodiments. The action interface (trajectory text vs. regression values) determines the failure surface, not the failing axis. Both policies retain closed-loop success under the composed recipe on the Orin, and open-loop metrics fail to certify either. We conclude with a practitioner decision procedure for quantizing VLA policies for edge deployment.

Index Terms—vision-language-action models, post-training quantization, closed-loop evaluation, edge deployment, Jetson Orin, autonomous driving, robot manipulation, activation outliers, group-wise quantization

I. Introduction

Vision–language–action (VLA) policies now drive both cars and robot arms, but their multi-billion parameter language backbones are built for datacenter GPUs, not the embedded accelerators that ship inside a vehicle or a robot. Post-training quantization (PTQ) is the standard bridge. The problem is how quantization is validated: almost always on in-distribution, open-loop accuracy—next-token or single-step error against a logged reference—

whereas the deployed policy runs closed-loop, where its own quantized outputs feed back into the world and errors compound over a rollout. We show this gap matters, in two embodiments, on real edge hardware.

We study PTQ for closed-loop success on the edge across two very different VLA policies deployed on the same NVIDIA Jetson AGX Orin: a driving policy (OpenDriveVLA [1], an autoregressive-text trajectory head on nuScenes, tested closed-loop in the NeuroNCAP [5] neural-rendering safety simulator) and a manipulation policy (OpenVLA-OFT [9], an L1-regression action head, tested closed-loop on LIBERO [10]). The stakes differ—a corrupted driving plan is a collision, a corrupted grasp is a dropped object—but the systems constraint (embedded latency, power, and shared memory) is shared, and so, we find, is the structure of how quantization fails.

- A cross-embodiment closed-loop study of VLA quantization on a single edge accelerator (Jetson AGX Orin), covering a driving policy (OpenDriveVLA / NeuroNCAP) and a manipulation policy (OpenVLA-OFT / LIBERO).
- Evidence that open-loop / in-distribution accuracy fails to certify quantized policies that closed-loop testing rejects—in both embodiments.
- A two-axis failure taxonomy (weight granularity under domain shift; activation outliers universally) with near-free fixes (group-wise scales; orthogonal rotation) that transfer across embodiments; the action interface sets the failure surface, not the axis.
- A unified on-Orin deployment comparison (closed-loop success, memory, latency/throughput, energy) and a decision procedure for quantizing VLAs for the edge.

II. Related Work

VLA policies. In driving, UniAD [2] and VAD [3] established query-based planning-oriented perception; OpenDriveVLA [1] attaches a Qwen2.5 language/action head that emits trajectories as text. In manipulation, OpenVLA [8] and its optimized fine-tuning recipe OpenVLA-OFT [9] pair a DINOv2+SigLIP vision tower and a Llama-2-7B backbone with an L1-regression action head. LLM weight quantization. RTN, GPTQ [11], and AWQ [12] choose quantization grids at group-wise granularity to bound weight-outlier damage, motivated by the outlier phenomenon in LLMs [13], [14], [17]. Activation quantization and rotation. Activation outliers are harder; incoherence-processing methods (QuaRot [15], SpinQuant [16]) insert an orthogonal rotation so quantization sees near-Gaussian operands. Closed-loop evaluation. NeuroNCAP [5] (NeuRAD [6] rendering) provides photorealistic closed-loop driving safety tests; LIBERO [10] provides closed-loop manipulation rollouts. Positioning. Prior VLA quantization work is manipulation-only and reports open-loop or in-distribution accuracy. To our knowledge this is the first study of VLA quantization for closed-loop success that spans two embodiments on a common edge accelerator, and the first to show the failure modes and their fixes transfer between them.

III. Background and Setup

A. Two policies, two action interfaces

The two policies span the design space that matters for this study—the action interface. OpenDriveVLA-0.5B places a Qwen2.5-0.5B language head (hidden 896, 24 layers) on UniAD stage-1 track+map perception and emits the trajectory as autoregressive text (168 backbone Linear layers, 357.8 M params quantized). OpenVLA-OFT pairs a fused DINOv2+SigLIP vision tower (1.22 B) and a Llama-2-7B backbone (6.48 B, 224 backbone Linear layers, the quantization target) with an L1-regression action head that outputs 8-action chunks. Text head vs. regression head is the contrast we exploit.

B. Two closed-loop benchmarks

For driving, NeuroNCAP re-renders six photorealistic cameras at the ego’s driven pose (frontal/side/ stationary collision scenarios) and we report generation coherence—the fraction of frames with a well-formed trajectory—because collision/NCAP is confounded by scene difficulty. For manipulation, LIBERO provides four suites (Spatial/Object/Goal/Long, 10 tasks each) scored by closed-loop task success over rollouts (step budgets 220/280/300/520; $\pm 4\%$ binomial CI at 100 rollouts, $\pm 2.5\%$ at 400).

C. Quantization methods

We use one shared formalism for both policies: symmetric round-to-nearest (RTN) weight quantization at a chosen granularity (per-output-channel or group-wise),

optional activation quantization, and optional orthogonal rotation for incoherence processing.

IV. Method

A. Unified quantization formalism

For a backbone linear $\mathbf{y} = W\mathbf{x}$ ($W \in \mathbb{R}^{d_{\text{out}} \times d_{\text{in}}}$), b -bit symmetric RTN over an index block $\mathcal{B} \subseteq \{1, \dots, d_{\text{in}}\}$ sets a scale $s_{\mathcal{B}} = \max_{j \in \mathcal{B}} |w_j| / (2^{b-1} - 1)$ and rounds $\hat{w}_i = s_{\mathcal{B}} \lfloor w_i / s_{\mathcal{B}} \rfloor$ (clipped to the grid). Granularity is the choice of $\mathcal{B}$: per-output-channel uses one block per row; group-wise uses one scale per g contiguous inputs, so a single weight outlier inflates the shared scale of only its own block rather than the whole row. Activation quantization applies the same rule per token to $\mathbf{x}$. Rotation inserts an orthogonal Q ($Q^\top Q = I$) on the input axis, $\mathbf{y} = W\mathbf{x} = (WQ^\top)(Q\mathbf{x})$, which is exact in full precision but lets quantization see the rotated, near-Gaussian operands $WQ^\top$ and $Q\mathbf{x}$; we use an orthonormal DCT for Q , which is data-free and defined for any dimension (unlike Hadamard, which requires powers of two).

B. On-device measurement methodology

Both policies are evaluated on an A100 (reference) and the AGX Orin (deployment) with identical quantization. For driving, we isolate device $\times$ quantization without porting UniAD’s custom CUDA perception ops by capture-replay: the fused planner input is quantization-independent, so we capture it from the A100 pipeline, transfer it, and replay it through the decoder-only planner on the Orin. For manipulation, LIBERO rollouts run directly on the Orin. Closed-loop success in both is measured with fake-quant weights (Sec. V-D details which systems numbers come from a real packed kernel).

V. Results

A. Driving VLA: open-loop is blind; granularity fixes closed-loop

On open-loop nuScenes L2, weight quantization is lossless to INT4 (Table I); the INT3/INT2 drop is generation collapse (malformed text), not graded planning. A controlled ego-ablation—zeroing the ego-state and 2 s history in the prompt, perception untouched—inflates L2 by $19\times$, showing the metric is $\approx$ ego-motion extrapolation and cannot see quantization damage.

Closed-loop tells a different story. In NeuroNCAP (14 scenes, 16 scenario instances, 10 paired seeds, real-camera NeuRAD rendering) we measure generation coherence—the fraction of frames on which the planner emits a well-formed (non-degenerate) trajectory. Under this deployment-domain shift, per-channel INT4 collapses ($\leq 15\%$ well-formed on 15/16, mean 3.7%) while group-wise INT4 recovers the bulk of FP16 coherence (mean 73.5% vs 76.0%, $\geq 85\%$ of FP16 on 13/16). The same per-channel INT4 was lossless in-distribution (Table I):

TABLE I

Driving, open-loop nuScenes-mini ($n=162$), ST-P3 L2 (m). Weight quantization is lossless to INT4; the ego-ablation reveals the metric is ego-extrapolation.

Setting	L2 (m)	well-formed
FP16	0.33	100%
INT8 / INT4 (per-channel)	0.33	100%
INT3	1.91	92%
INT2	6.45	0%
FP16, ego-state+history zeroed	6.38 (19 $\times$)	100%

TABLE II

Driving, closed-loop NeuroNCAP generation coherence (%), 14 scenes / 16 scenario instances, 10 paired seeds. Per-channel INT4 collapses under domain shift; group-wise recovers.

scene / scenario	FP16	INT4 per-ch.	INT4 g128
0099 stationary	60.6	0.0	30.0
0101 stationary	75.0	14.3	46.7
0103 frontal	74.3	0.0	72.9
0106 frontal	66.9	0.0	78.7
0108 side	85.7	0.0	91.6
0108 stationary	72.4	0.0	74.7
0110 frontal	70.7	0.0	74.3
0278 side	75.3	0.0	67.9
0278 stationary	79.7	0.0	83.5
0331 stationary	83.2	5.0	97.5
0346 frontal	83.1	15.6	86.9
0783 stationary	72.4	0.0	73.1
0796 stationary	72.4	8.6	57.9
0921 side	90.0	14.2	90.4
0923 frontal	74.6	0.8	72.1
0966 stationary	79.4	0.0	78.3
mean	76.0	3.7	73.5

the failure is a quantization $\times$ domain-shift interaction governed by scale granularity.

Collision/NCAP is a confounded proxy here: on several scenes every configuration including FP16 collides 10/10 under the OOD renderer, so collisions cannot isolate the quantization effect; we lead with coherence.

B. Manipulation VLA: activations are the wall; rotation fixes closed-loop

For the manipulation policy (OpenVLA-OFT, Llama-2-7B backbone, L1-regression action head), the mirror-image result holds on LIBERO closed-loop success (Table III). Weight-only quantization is easy—naive per-channel INT4 is near-lossless (90.75% vs 90.25% bf16). But quantizing activations to 4 bits collapses the policy to 0% success: a single backbone channel (layers.1.mlp.down_proj) carries a magnitude $\sim 10^5 \times$ the median, and a shared activation scale set by that spike zeros out every other channel. An orthogonal rotation $y = (WQ^T)(Qx)$ that spreads this energy restores W4A4 to $\geq 98\%$ retention.

The rotation’s effect is quantitative: the median activation-outlier ratio drops $18.0 \rightarrow 1.6$ (11 $\times$) and the worst layer $105,928 \rightarrow 6.2$ (17,000 $\times$), cutting the median INT4 per-tensor error from 96.5% to 26.1%. A

TABLE III

Manipulation, LIBERO closed-loop success (% over 400 rollouts, all 4 suites; bf16 baseline 90.25%). Weights quantize freely; activation-4bit collapses the policy to 0%; an orthogonal rotation restores it.

Config	act. bits	success	retention
bf16 baseline	16	90.25%	—
Naive W4 (per-channel)	16	90.75%	100.6%
Naive W3 (per-channel)	16	89.2%	98.8%
Naive W4A4	4	0.0%	0%
Rotated W4A4 (DCT)	4	89.8%	99.5%
Rotated W4A4 (Hadamard)	4	88.8%	98.4%
Rotated W3A8 (DCT)	8	89.0%	98.6%

TABLE IV

Crossover on the driving policy (decoder-only replay of 162 real inputs, in-distribution). Activation-4bit alone collapses it; the manipulation-side fix (DCT rotation) restores it—the fix transfers. FP16 base rate: 42/162 legitimately-stationary all-zero plans.

Config	ST-P3 L2 avg (m)	all-zero plans	$ xy > 50$ m
FP16 (ref)	0.595	42 (base)	0
W16A4, no rotation	27.79 (47 $\times$)	107	7
W16A4 + DCT	0.646	42	0
W4A4, no rotation	9.34	127	3
W4A4 + DCT	0.611	41	0

component-wise analysis further shows the three modules have different floors: the vision tower is free (INT8/FP8 lossless), the backbone is the main win (DCT-rotated W4A4/W3A8 lossless), and the L1-regression head is fragile (INT4 drops to 70%, INT3 to 50%)—so it must stay $\geq$ INT8. Composing these (INT8 vision + DCT-W3A8 backbone + INT8 head) yields 74% compression at no measurable accuracy loss (89.5% vs 88.0% bf16, all-suite; 15.4 $\rightarrow$ 4.0 GB).

C. Crossover: the fixes transfer across embodiments

The two fixes are not embodiment-specific. We test the manipulation-side fix (orthogonal rotation to tame activation outliers) on the driving policy. Quantizing OpenDriveVLA’s activations to 4 bits—weights left at FP16—collapses the planner in-distribution (Table IV), exactly where weight-only INT4 was lossless: the massive-activation wall transfers from the 7B manipulation backbone to the 0.5B driving backbone. A single data-free DCT rotation restores W4A4 to FP16-level accuracy and returns the degenerate-plan count to its base rate.

Two observations complete the taxonomy. First, the axes are distinct triggers: weight-scale granularity fails only under domain shift (Table II), while activation outliers fail universally and in-distribution (Table IV). Second, the action interface sets the failure surface: the driving policy’s activation collapse produced well-formed but degenerate trajectory content (0 malformed text), whereas its domain-shift weight collapse produced malformed trajectory syntax. Both freeze the robot; a syntactic monitor catches one and not the other.

D. Unified on-Orin deployment

Both policies were deployed on the same NVIDIA Jetson AGX Orin 64 GB (JetPack 6 / L4T R36.4.7, CUDA cap 8.7, PyTorch 2.8). Table V places them side by side: the quantized recipe retains closed-loop performance on-device in both embodiments (96.7% and 97.5% of the full-precision reference), while cutting the weight footprint several-fold.

Two honest systems notes. First, both closed-loop-success numbers were produced with fake-quant (weights dequantized to the compute dtype), so neither yet reflects a packed-INT4 kernel; the driving planner’s static footprint win is real ($0.716 \rightarrow 0.185$ GB packed, $3.9\times$) but its latency is precision-flat (≈ 4.0 s for 1024-token context $\rightarrow 60$ tokens), and the manipulation policy’s fake-quant latency is likewise near its bf16 value (~ 330 vs 320 ms/query). Second, a real packed-INT4 kernel does exist for the manipulation backbone (GGUF Q4_K_M via llama.cpp, sm_87): 3.8 GiB backbone ($3.3\times$ memory vs 12.5 GiB bf16), 903/28 tok/s prefill/decode—but it was benchmarked for memory/throughput only, not run closed-loop. The equivalent driving kernel is future work. What is solid on-device is accuracy fidelity: the driving planner’s Orin trajectories match the A100 to ≈ 1.1 cm mean (FP16) / 1.5 cm (group-INT4), and the manipulation policy stays within its $\pm 4\%$ rollout confidence interval of the server. This is the throughput/power/shared-budget case for quantizing a VLA even when it fits: on production compute the policy shares memory and bandwidth with FP32 perception and the rest of the stack, and the quantized recipe frees both while holding closed-loop success.

VI. Discussion: a decision procedure

The cross-embodiment picture yields a concrete recipe for quantizing a VLA policy for the edge. (1) Always use group-wise weight scales. They cost ≈ 0.1 extra bits/weight and prevent the domain-shift-triggered weight collapse (Table II); per-channel is adequate in-distribution but unsafe closed-loop. (2) If activations are quantized, rotate first. An orthogonal (DCT) rotation converts the otherwise-fatal activation-4bit collapse into a $\geq 98\%$ -retention recipe in both embodiments (Tables III, IV). (3) Certify closed-loop, not open-loop. Open-loop/in-distribution accuracy certified quantized policies that closed-loop testing rejected in both embodiments. (4) Match the runtime monitor to the action interface. The failure surface depends on the head: a text head fails as malformed trajectory syntax (catchable by a parser), a regression head as well-formed but degenerate values (not)—so a deployed policy needs a monitor keyed to its output type, not a generic one.

VII. Limitations

The driving closed-loop test uses an out-of-distribution neural renderer, so its absolute NCAP numbers are

not in-distribution-comparable and our driving claims are relative (group $\gg$ per-channel on coherence). Each embodiment is one checkpoint on one benchmark. Closed-loop success in both is measured with fake-quant weights; a packed-INT4 kernel is deployed for the manipulation backbone (memory/throughput only) and is future work for driving, so we do not report a measured real-kernel closed-loop speedup. Energy-per-inference was not measured (pending on-board tegrastats). The crossover establishes that the two fixes transfer, but we do not exhaustively sweep every axis $\times$ embodiment cell.

VIII. Conclusion

On a single edge accelerator and two very different VLA policies, quantization for deployment must be certified closed-loop and on-device: open-loop accuracy hid failures that closed-loop testing exposed in both embodiments. Those failures follow a common two-axis structure—weight-scale granularity (fails under domain shift) and activation outliers (fails universally)—each with a near-free, transferable fix (group-wise scales; orthogonal rotation), while the action interface sets only the failure surface. The resulting recipe retains closed-loop success on the Jetson Orin in both driving and manipulation, and the accompanying decision procedure applies to any VLA headed for embedded compute.

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TABLE V
 Unified on-Orin deployment (AGX Orin 64 GB). Both VLA policies, same board; closed-loop success measured with fake-quant (dequantized) weights in both cases. The quantized recipe retains closed-loop performance in both embodiments. Packed-INT4 kernels (for the true memory/latency win) exist for manipulation and are pending for driving; neither closed-loop success number is yet from a packed kernel.

Policy (embodiment)	Backbone	Closed-loop metric	Full-prec.	Quantized recipe	On-Orin (retention)	Packed-INT4
OpenDriveVLA (driving)	Qwen2.5-0.5B	NeuroNCAP coherence %	76.0	W4 group-128	73.5 (96.7%)	pending
OpenVLA-OFT (manip.)	Llama-2-7B	LIBERO-Spatial succ. %	88.2	W4A4 + DCT	86.0 (97.5%)	GGUF Q4_K_M (100%)